for $n=4$

1. a)

$$P_2 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \end{bmatrix} \begin{matrix} \Rightarrow \\ \Rightarrow \\ \Rightarrow \\ \Rightarrow \end{matrix}$$

$\pi = \pi P$

$$\Rightarrow \begin{bmatrix} \pi_1 & \pi_2 & \pi_3 & \pi_4 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}\pi_4 & \pi_1 + \frac{1}{2}\pi_4 & \pi_2 & \pi_3 \end{bmatrix}$$

$\pi_1 = \frac{1}{2}\pi_4$

$\pi_2 = \pi_1 + \frac{1}{2}\pi_4 = \frac{1}{2}\pi_4 + \frac{1}{2}\pi_4 = \pi_4$

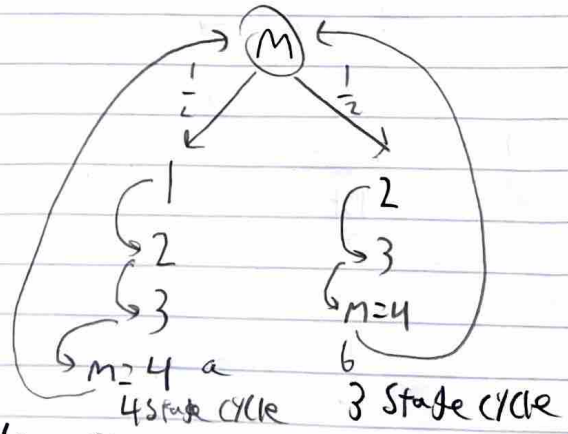
$\pi_3 = \pi_2 = \pi_4$

$\pi_4 = \pi_3 = \pi_4$

$\pi_1 + \pi_2 + \pi_3 + \pi_4 = 1$

$\frac{1}{2}\pi_4 + \pi_4 + \pi_4 + \pi_4 = \frac{7}{2}\pi_4 = 1 \Rightarrow \pi_4 = \frac{2}{7} \quad \pi_1 = \frac{1}{7}$

$$\Rightarrow \pi = \begin{bmatrix} \frac{1}{7} \\ \frac{2}{7} \\ \frac{2}{7} \\ \frac{2}{7} \end{bmatrix}$$



$n=9$ does not return to 1 because there is no combination of 3 or 4 stage cycles that works, i.e. $9 = 4a + 3b + 1$ only works for 1st initial and 2 4 stage cycles, while $n \geq 10$ can have 2 3 stage cycles and 1 4 stage cycle, and all n greater work for $n = 4a + 3b + 1$

b) $P_{m=2} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \end{bmatrix} \rightarrow P_{m=3} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{bmatrix}$ $m=4$

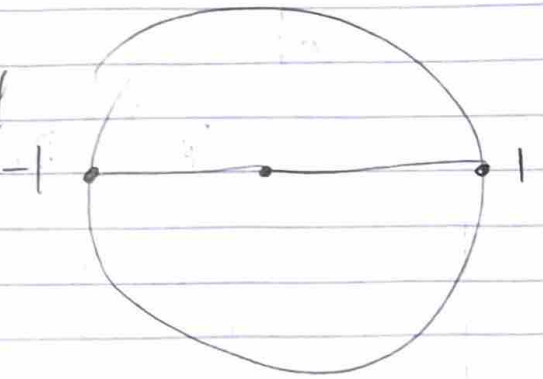
$m=3$

$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$

aperiodic greatest divisor is 1
Periodic greatest divisor is 2

for M = odd # it is aperiodic since the greatest common divisor of M is 1, for even #'d M, ther are periodic with period 2:

$\lambda = 1$ for $M=4$



c) Table found using matlab

2. Given: m teams, r is ranking

$\lambda r_i = \sum_j a_{ij} r_j \Rightarrow$ if $\lambda = 1$ $r = Ar \Rightarrow$ stationary distribution

in a game

team i scores S, team j scores T

i recieves $h\left(\frac{S+1}{S+T+2}\right)$ w/ $h(x) = \frac{1}{2} + \frac{1}{2}\,\mathrm{sign}\left(x-\frac{1}{2}\right)\sqrt{|2x-1|}$

$\Pi_{i,j} \in (0,1)$ $\Pi_{ij} + \Pi_{ji} = 1$ $w \geq 0$

$$a_{ij} = \frac{\sum h_{ij} + w\,\Pi_{ij}}{n_i + w(m-1)}$$

a) $\lambda r_i = \sum_j a_{ij} \cdot r_j$

if $\lambda \geq \max_i \sum_j a_{ij}$

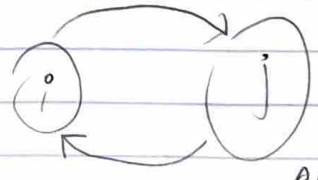
then $\lambda r_i \geq \sum_j a_{ij} \cdot r_j$, and λ would not be an eigenvalue

if $\lambda \leq \min_i \sum_j a_{ij}$

then $\lambda r_i \leq \sum_j a_{ij} r_j$, and λ would not be an eigen value

if $w = 0$

$$a_{ij} = \frac{\sum_{\text{games } i \text{ vs } j} h_{ij}}{n_i}$$



2 team block since when 1 team plays the other, the other team plays them

$$\Rightarrow A = \begin{bmatrix} \sum h_{ij}/n_i & 0 \\ 0 & \sum j_i/n_i \end{bmatrix}$$

$$|A - I\lambda| = \begin{vmatrix} a_{ij} - \lambda & 0 \\ 0 & a_{ji} - \lambda \end{vmatrix} = (a_{ij} - \lambda)(a_{ji} - \lambda) = 0$$

$\Rightarrow \lambda = a_{ij}$ or a_{ji}

a) this is not unique since since

$$a_{ij} = \frac{\sum h_{ij}}{n_i} = a_{ji} = \frac{\sum h_{ji}}{n_i} \quad \text{since } h = \left(\frac{s+1}{s+r+2}\right)$$

b) $w > 0 \quad m \geq 3 \; A^2 > 0$

Perron Frobenius

If $A = P^T$

then $q_{n+1}^T = A q_n^T$

for P irreducible $\lambda = 1$

P aperiodic $|\lambda_i| \leq 1$ strictly

for $w > 0 \quad m \geq 3 \quad A^2 > 0$

$h(x) = \frac{1}{2} + \frac{1}{2}\,\text{sign}(x - \frac{1}{2})\sqrt{|2x-1|}$ is ≥ 0 for $h\left(\frac{s+1}{s+r+2}\right)$ since $\frac{s+1}{s+r+2} < 1$

$\Rightarrow a_{ij} > 0$ since $\frac{\sum h_{ij} + w\Pi_{ij}}{n_i + w(m-1)} > 0$ since $\sum h_{ij} > 0$, $w\Pi_{ij} \geq 0$, $n_i + w(m-1) > 0$

$\Rightarrow A^2 > 0$

Perron Frobenius Guarantees

$r > 0$ since $A > 0$

(eigenvector is greater than 0)

since $r > 0$

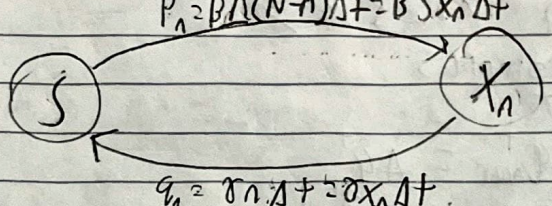
all power iterations of $r > 0$

3. Given: X_n infected out of N total hens

$p_n = \beta n(N-n)\Delta t$, $q_n = \gamma n \Delta t$, and 0 is absorbing

$R_0 = \frac{\beta N}{\gamma}$ $\quad T = \min\{n : X_n \in \{0, k\}\}$ w/ $k < N$

$P_n = \beta S X_n \Delta t$



$S + X_n = N$

$S = N - n$ $\quad X_n = n$

a) $h_n = P_n(X_T = k) = \frac{\sum_{j=0}^{n-1} \rho_j}{\sum_{j=0}^{k-1} \rho_j}$ $\quad \rho_0 = 1$ $\quad \rho_j = \prod_{k=1}^{j} \frac{q_k}{p_k}$

$= \frac{\rho_0 + \rho_1 + \rho_2 + \cdots \rho_{n-1}}{\rho_0 + \rho_1 + \rho_2 + \cdots \rho_{n-1} + \rho_n + \cdots \rho_{k-1}} = \frac{1 + \frac{q_1}{p_1} + \frac{q_2}{p_2}\frac{q_1}{p_1} + \cdots \frac{q_{n-1}}{p_{n-1}} \cdot \frac{q_{n-2}}{p_{n-2}} \cdots \frac{q_1}{p_1}}{1 + \frac{q_1}{p_1} + \frac{q_2}{p_2}\frac{q_1}{p_1} + \cdots \frac{q_{n-1}}{p_{n-1}}\frac{q_{n-2}}{p_{n-2}} \cdots \frac{q_1}{p_1} + \cdots \frac{q_{n+1}}{p_{n+1}} \cdots \frac{q_1}{p_1}}$

$p_n(h_{n+1} - h_n) = q_n(h_n - h_{n-1}) \Rightarrow \beta(h_{n+1} - h_n)(N - (h_{n+1} - h_n))\Delta t = \gamma(h_n - h_{n-1})\Delta t$

$\Rightarrow N(h_{n+1} - h_n) - (h_{n+1} - h_n)^2 = \frac{\gamma}{\beta}(h_n - h_{n-1})$

Assume:

$h_{n+1} - h_n = \frac{\sum_{j=0}^{n} \rho_j}{\sum_{j=0}^{k-1} \rho_j} - \frac{\sum_{j=0}^{n-1} \rho_j}{\sum_{j=0}^{k-1} \rho_j} = \frac{\rho_n}{\sum_{j=0}^{k-1} \rho_j}$

$h_n - h_{n-1} = \frac{\sum_{j=0}^{n-1} \rho_j}{\sum_{j=0}^{k-1} \rho_j} - \frac{\sum_{j=0}^{n-2} \rho_j}{\sum_{j=0}^{k-1} \rho_j} = \frac{\rho_{n-1}}{\sum_{j=0}^{k-1} \rho_j}$

$\Rightarrow N\left(\frac{\rho_n}{\sum_{j=0}^{k-1} \rho_j}\right) - \frac{\rho_n^2}{\left(\sum_{j=0}^{k-1} \rho_j\right)^2} = N \frac{\rho_n \sum_{j=0}^{k-1} \rho_j - \rho_n^2}{\left(\sum_{j=0}^{k-1} \rho_j\right)^2}$

3 a)

$$\frac{N\rho_n \sum_{j=0}^{k-1}\rho_j - \rho_n^2}{\left(\sum_{j=0}^{k-1}\rho_j\right)^2} = \frac{\partial}{\beta}\frac{\rho_{n-1}\sum_{j=0}^{k-1}\rho_j}{\left(\sum_{j=0}^{k-1}\rho_j\right)^2}$$

$$\Rightarrow \frac{N\rho_n \sum_{j=0}^{k-1}\rho_j - \rho_n^2 - \frac{\partial}{\beta}\rho_{n-1}\sum_{j=0}^{k-1}\rho_j}{\left(\sum_{j=0}^{k-1}\rho_j\right)^2} = 0$$

$$\Rightarrow N\rho_n \sum_{j=0}^{k-1}\rho_j - \rho_n^2 - \frac{\partial}{\beta}\rho_{n-1}\sum_{j=0}^{k-1}\rho_j = 0$$

ρ_n

$$N\sum_{j=0}^{k-1}\rho_j - \rho_n - \frac{\partial}{\beta}\frac{\rho_{n-1}}{\rho_n}\sum_{j=0}^{k-1}\rho_j = 0$$

ρ_n

$$\frac{p_n}{q_n} = \frac{\beta}{\partial}(N-n)$$

$$\Rightarrow N\sum_{j=0}^{k-1}\rho_j - \rho_n - (N-n)\sum_{j=0}^{k-1}\rho_j = 0$$

$$\Rightarrow n\sum_{j=0}^{k-1}\rho_j = \rho_n$$

$$\boxed{n = \frac{\rho_n}{\sum_{j=0}^{k-1}\rho_j} = h_{n+1} - h_n}$$

shows how $h_n = \frac{\sum_{j=0}^{n-1}\rho_j}{\sum_{j=0}^{k-1}\rho_j}$

is soln to

$$p(h_{n+1} - h_n) = q_n(h_n - h_{n-1})$$

$1 - p_n - q_n$ drops out

$$p_j = \prod_{k=1}^{j} q_k / p_k = \prod_{k=1}^{j} \frac{\gamma(k)\Delta t}{\beta(k)(N-k)\Delta t} = \prod_{k=1}^{j} \frac{\gamma}{\beta} \cdot \frac{1}{N-k} = \prod_{k=1}^{j} \frac{N}{R_0(N-k)}$$

$R_0 = \beta N/\gamma$ $\quad \frac{1}{R_0} = \frac{\gamma}{\beta N}$ $\quad \frac{N}{R_0} = \frac{\gamma}{\beta}$

3. b) $p_j = \prod_{k=1}^{j} \frac{N}{R_0(N-k)}$ $\quad$ let $N \to \infty$ k fixed

$p_j = \prod_{k=1}^{j} \frac{N}{R_0(N-k)}$ $\quad$ then $k \to \infty$ $\quad h_1 \to \max(0, 1 - 1/R_0)$

$\lim_{N\to\infty} \frac{N}{R_0(N-k)}$ $\quad \frac{\lim_{N\to\infty} N = \infty}{\lim_{N\to\infty} R_0(N-k) = \infty}$ L'Hopitals Rule

$$\Rightarrow \lim_{N\to\infty} \frac{N}{R_0(N-k)} = \lim_{N\to\infty} \frac{1}{R_0}$$

$$\lim_{k\to\infty} \frac{N}{R_0(N-k)} = 0$$

$$h_1 = \frac{\sum_{j=0}^{k-1} p_j = p_0 + p_1 + \cdots + p_{k-1}}{\sum_{j=0}^{k-1} p_j} = \frac{1 + \cdots + p_{k-1}}{p_0 + p_1 + p_2 + \cdots + p_{k-1}} = \frac{1}{1 + \frac{N}{R_0(N-1)} + \frac{N}{R_0(N-2)} \cdot \frac{N}{R_0(N-1)} + \left(\frac{N}{R_0}\right)^{k-1} \frac{1}{(N-(k-1))!}}$$

$$\lim_{N\to\infty} h_1 = \frac{1}{1 - \frac{1}{R_0}(k-1)} = \lim_{k\to\infty} \frac{1}{1 - \frac{1}{R_0}(k-1)}$$

3 c) In MATLAB

d)

$$P = \begin{array}{c} \begin{matrix} n=1 & n=2 & n=3 & \cdots & & n=N \end{matrix} \\ \begin{bmatrix} 1-p_1 & p_1 & 0 & \cdots & & 0 \\ q_2 & 1-p_2-q_2 & p_2 & 0 & \cdots & \\ 0 & q_3 & 1-p_3-q_3 & p_3 & & \\ \vdots & & \ddots & \ddots & \ddots & \\ & & & & p_{N-1} & \\ & & & q_N & 1-q_N & \\ 0 & & & & & \end{bmatrix} \end{array} \begin{matrix} e_1 \\ e_2 \\ e_3 \\ \vdots \\ e_{100} \end{matrix}$$

$E[X] = \sum x_i P(x_i)$

$\Rightarrow e_1 = 1(1-p_1) + 2(p_1) = 1 - p_1 + 2p_1 = 1 + p_1 = 1 + .0025 = \boxed{1.0025 = e_1}$

$e_2 = q_2 \cdot 1 + (1-p_2-q_2)(2) + 3(p_2) = 1 + p_2 - q_2 = e_2$

$e_3 = q_3 \cdot 2 + (1-p_3-q_3)3 + 4(p_3) = 1 + p_3 - q_3 = e_3$

$\vdots$

$e_{N-1} = 1 + p_{N-1} - q_{N-1} \qquad 1 + p_{N-1} - q_{N-1} = e_{N-1}$

$e_N = q_N(N-1) + (1-q_N)(N) = 1 - q_N = e_N$

$e_N \Delta t$ is independent of Δt since Δt is applied equally through matrix